

A consumer-relative reading of entropy-monotonic geometric flows

Andrew H. Bond
San José State University

Speculative note, 2026-08-23 — **a question for a geometer, not a result**

Abstract

Observation Theory (OT) treats reliability and distortion as properties of a *consumer* through a *read operator* $P_C \succeq 0$ that keeps only the distinctions the consumer can make, inducing a quotient of the underlying distinguishability geometry. Ricci flow is the gradient flow of Perelman’s entropy, and its analysis is driven by a Bakry–Émery/Fisher-information substrate that OT also lives on. This note asks, honestly, whether OT’s *read-operator quotient* says anything non-trivial about entropy-monotonic geometric flows: specifically, whether coarse-graining a flow by P_C carries a *consumer-relative* entropy with a monotone “discarded-distinguishability” gap. I state the observation precisely, situate it against known results (Perelman; O’Neill submersions; Bakry–Émery; Lott–Villani–Sturm; Otto), and give the ways an expert can falsify or dismiss it in minutes. I claim no theorem; OT’s contribution here, if any, is vocabulary and a reframing, not a new technique. I write at the edge of my competence and expect correction.

1 Setup

On (M^n, g) , Ricci flow is $\partial_t g = -2 \operatorname{Ric}(g)$. Perelman’s $\mathcal{F}(g, f) = \int_M (R + |\nabla f|^2) e^{-f} dV_g$, coupled to a backward heat equation for f preserving $\int e^{-f} dV$, is monotone:

$$\frac{d}{dt} \mathcal{F} = 2 \int_M |\operatorname{Ric} + \nabla^2 f|^2 e^{-f} dV \geq 0.$$

The integrand is $|\operatorname{Ric}_f|^2$ for the Bakry–Émery tensor $\operatorname{Ric}_f = \operatorname{Ric} + \nabla^2 f$, so Perelman’s weight f is a Bakry–Émery weight and the flow is the gradient flow of an entropy. The scale-invariant $\mathcal{W}$ -entropy is likewise monotone, with critical points the gradient shrinking solitons.

2 OT’s read operator as coarse-graining

Read a point of M as (the sufficient statistics of) a probability model, with the Fisher–Rao metric as the distinguishability geometry. A read operator P_C that retains only some distinctions is a projection onto a sub-distribution of observables; geometrically it is a smooth surjection — a coarse-graining $\pi : (M, g) \rightarrow (N, h)$, the consumer’s world $N = M/\sim$ being the quotient by the fibers $\pi^{-1}(y)$ it cannot separate. When π is a Riemannian submersion, P_C is orthogonal projection onto the horizontal distribution.

3 Three facts that make the question well-posed

- (O) **Coarse-graining does not lose Ricci (O’Neill).** For a Riemannian submersion $\pi : (M, g) \rightarrow (N, h)$, O’Neill’s equations give $\operatorname{Ric}_h = (\text{horizontal } \operatorname{Ric}_g) + (\text{non-negative } A\text{-tensor corrections})$;

the base is *more* positively curved. The consumer’s quotient inherits curvature *plus* a correction.

- (D) **The flow descends under symmetry.** Ricci flow does not generally preserve a submersion, but for metrics invariant under an isometry group G (homogeneous / cohomogeneity-one), it does: $g(t)$ stays invariant and descends to a flow $h(t)$ on $N = M/G$. So a *consumer-relative Ricci flow* is well-defined at least when P_C quotients by a symmetry.
- (P) **Entropy contracts under coarse-graining (DPI).** Relative entropy is non-increasing under a Markov/coarse-graining map: $\text{Ent}(\pi_{\#}\mu\|\pi_{\#}\nu) \leq \text{Ent}(\mu\|\nu)$. Perelman’s $\mathcal{F}, \mathcal{W}$ are entropies; a consumer reads their pushforward.

4 The question

Combining (O), (D), (P): let π quotient by a symmetry so the flow descends, and let $\mathcal{W}_C(h, \bar{f}, \tau)$ be Perelman’s $\mathcal{W}$ computed on the quotient (N, h) .

Question 1. *Along the descended (consumer-relative) flow, is $\mathcal{W}_C$ monotone, and is the gap $\Delta_C(t) = \mathcal{W}(g(t)) - \mathcal{W}_C(h(t))$ monotone (say non-increasing)?*

Conjecture 1 (consumer-discarded distinguishability). *For quotients by a compact isometry group with totally geodesic fibers, the gap $\Delta_C(t) \geq 0$ is monotone non-increasing along Ricci flow: the distinguishability the consumer cannot read is dissipated monotonically by the flow, at a rate controlled by the O’Neill A -tensor and the fiber Bakry–Émery Ricci curvature.*

The OT reading: Δ_C is the reliability/entropy in the fibers — exactly what P_C quotients out — and its monotonicity would be a geometric data-processing inequality *coupled to* the flow, indexed by the consumer.

5 Honest situating

Every ingredient is classical and active, which bounds the novelty sharply: homogeneous and cohomogeneity-one Ricci flow are well developed; O’Neill submersion curvature is 1966; entropy DPI is elementary; and the synthetic picture (Lott–Villani–Sturm $\text{CD}(K, N)$, Otto calculus, Bakry–Émery) already couples Ricci, entropy, and optimal transport, with substantial theory on quotients and measured-Gromov–Hausdorff stability. So Question 1 is very possibly (i) an immediate corollary of Bakry–Émery + O’Neill + DPI — in which case OT contributes only a clean *name* for a known gap; or (ii) false because the O’Neill correction enters the $\mathcal{W}$ -monotonicity slot with an uncontrolled sign; or (iii) vacuous outside the symmetric case, where the flow need not descend. I do not know which. What OT genuinely adds is *not* a technique but a viewpoint: “what the consumer reads” as the invariant object, and the discarded-distinguishability gap as the quantity to track — a reframing that is only worth anything if it makes an otherwise-opaque statement obvious, and worthless otherwise.

6 How to dismiss or sharpen this in five minutes

An expert can grade it fast: (a) Is Δ_C monotonicity immediate from Ni’s/Perelman’s entropy formula applied to the quotient with the Bakry–Émery weight induced by the fiber volume? If yes, it is known — please name the reference. (b) Does the O’Neill A -tensor term have a definite

sign in $d\mathcal{W}_C/dt$? If not, Conjecture 1 is false as stated. (c) Is there a homogeneous example (e.g., a Berger-sphere collapse $S^3 \rightarrow S^2$ under Ricci flow) where $\Delta_C(t)$ can be computed explicitly and checked? That example alone would confirm or kill it.

Non-applicability, stated plainly. OT’s operational apparatus — witnessed certificates, false-clear rates, the sealed-measurement discipline — does *not* apply here: Ricci flow is a deductive problem with no consumer to serve and no outcome to measure. Only the information-geometric core of OT (the read-operator quotient of a Fisher/entropy geometry) is even a candidate, and only in the speculative sense above.

References

- [1] G. Perelman, “The entropy formula for the Ricci flow and its geometric applications,” arXiv:math/0211159, 2002.
- [2] R. Hamilton, “Three-manifolds with positive Ricci curvature,” J. Differential Geom., 1982.
- [3] B. O’Neill, “The fundamental equations of a submersion,” Michigan Math. J., 1966.
- [4] D. Bakry, M. Émery, “Diffusions hypercontractives,” Séminaire de Probabilités XIX, 1985.
- [5] J. Lott, C. Villani, “Ricci curvature for metric-measure spaces via optimal transport,” Ann. of Math., 2009.
- [6] K.-T. Sturm, “On the geometry of metric measure spaces I/II,” Acta Math., 2006.
- [7] F. Otto, “The geometry of dissipative evolution equations: the porous medium equation,” Comm. PDE, 2001.
- [8] P. Topping, *Lectures on the Ricci Flow*, LMS Lecture Notes, 2006.
- [9] S. Amari, *Information Geometry and Its Applications*, Springer, 2016.
- [10] A. H. Bond, “Tradeoffs Between Rate and Conditional Content with Encoder-Observed Context,” manuscript, 2026; preliminary record Zenodo, doi:10.5281/zenodo.21776291.